Publication List
Paul M. Chichura

Last updated: May 2026

Metrics. I have authored 33 publications, including two first-author publications. These publications have received 951 citations (H-Index: 16), according to NASA ADS.

Journal abbreviations. A&A: Astronomy & Astrophysics; ApJ: The Astrophysical Journal; ApJL: The Astrophysical Journal Letters; JCAP: Journal of Cosmology and Astroparticle Physics; MNRAS: Monthly Notices of the Royal Astronomical Society; PhRvD: Physical Review D; PhRvL: Physical Review Letters.

Note on Authorship. I am a member of the South Pole Telescope (SPT-3G) and Event Horizon Telescope (EHT) collaborations, which follow authorship policies wherein all contributing members are credited. I list all publications on which I am listed as a co-author, and I separate my first-author publications to highlight my primary individual contributions.

(1) First-Author Publications

[1] **Chichura, P. M.**, A. Rahlin, A. J. Anderson, et al. 2025. "Pointing Accuracy Improvements for the South Pole Telescope with Machine Learning." *Journal of Astronomical Instrumentation* 14 (January): 2550001. DOI: 10.1142/S2251171725500011. arXiv:2412.15167.

[2] **Chichura, P. M.**, A. Foster, C. Patel, et al. 2022. "Asteroid Measurements at Millimeter Wavelengths with the South Pole Telescope." *ApJ* 936, no. 2 (September): 173. DOI: 10.3847/1538-4357/ac89ec. arXiv:2202.01406.

(2) All Publications

[3] Archipley, M., A. Hryciuk, L. E. Bleem, et al. 2026. "Millimeter-wave observations of Euclid Deep Field South using the South Pole Telescope: A data release of temperature maps and catalogs." *A&A* 706 (January): A17. DOI: 10.1051/0004-6361/202555798. arXiv:2506.00298.

[4] Bernshteyn, V., N. S. Conroy, M. Bauböck, et al. 2026. "Ring Asymmetry and Spin in M87*." *ApJ* 1000, no. 2 (April): 231. DOI: 10.3847/1538-4357/ae34af. arXiv:2601.00394.

[5] Camphuis, E., W. Quan, L. Balkenhol, et al. 2026. "SPT-3G D1: CMB temperature and polarization power spectra and cosmology from 2019 and 2020 observations of the SPT-3G main field." *PhRvD* 113, no. 8 (April): 083504. DOI: 10.1103/7wt3-9v2y. arXiv:2506.20707.

[6] Chaubal, P., N. Huang, C. L. Reichardt, et al. 2026. "SPT-3G D1: A Measurement of Secondary Cosmic Microwave Background Anisotropy Power." *arXiv e-prints* (January): arXiv:2601.20551. DOI: 10.48550/arXiv.2601.20551. arXiv:2601.20551.

[7] de Haan, T., M. Archipley, N. Huang, et al. 2026. "Characterization of the Polarization Beam Response of SPT-3G Using Point Sources." *arXiv e-prints* (February): arXiv:2602.06334. DOI: 10.48550/arXiv.2602.06334. arXiv:2602.06334.

[8] Georgiev, B., P. Tiede, S. D. von Fellenberg, et al. 2026. "Locating the missing large-scale emission in the jet of M87* with short EHT baselines." *A&A* 709 (April): A26. DOI: 10.1051/0004-6361/202557013. arXiv:2601.13356.

[9] Maniyar, A. S., F. Bianchini, W. L. K. Wu, et al. 2026. "SPT-3G D1: Compton-y maps using data from the SPT-3G and Planck surveys." *arXiv e-prints* (February): arXiv:2602.11279. DOI: 10.48550/arXiv.2602.11279. arXiv:2602.11279.

[10] Qu, F. J., F. Ge, W. L. K. Wu, et al. 2026. "Unified and Consistent Structure Growth Measurements from Joint ACT, SPT, and Planck CMB Lensing." *PhRvL* 136, no. 2 (January): 021001. DOI: 10.1103/k5yr-3h6d. arXiv:2504.20038.

[11] Quan, W., E. Camphuis, C. Daley, et al. 2026. "SPT-3G D1: Maps of the millimeter-wave sky from 2019 and 2020 observations of the SPT-3G Main field." *arXiv e-prints* (March): arXiv:2603.20163. DOI: 10.48550/arXiv.2603.20163. arXiv:2603.20163.

[12] Raghunathan, S., P. A. R. Ade, D. Anbajagane, et al. 2026. "Measurement of the Full Shape of the Thermal SunyaevZel'dovich Power Spectrum from the South Pole Telescope and HerschelSPIRE Observations." *ApJL* 1002, no. 1 (May): L22. DOI: 10.3847/2041-8213/ae5c07. arXiv:2602.10107.

[13] Saurabh, H. Müller, S. D. von Fellenberg, et al. 2026. "Probing jet base emission of M87* with the 2021 Event Horizon Telescope observations." *A&A* 706 (January): A27. DOI: 10.1051/0004-6361/202557022. arXiv:2512.08970.

[14] Tandoi, C., A. Foster, T. J. Maccarone, et al. 2026. "Millimeter-wave Detections of Symbiotic Stars in SPT and ACT Data." *arXiv e-prints* (May): arXiv:2605.01022. DOI: 10.48550/arXiv.2605.01022. arXiv:2605.01022.

[15] Wan, Y., J. D. Vieira, **Chichura, P. M.**, et al. 2026. "Detection of Millimeter-wavelength Flares from Two Accreting White Dwarf Systems in the SPT-3G Galactic Plane Survey." *ApJ* 997, no. 2 (February): 133. DOI: 10.3847/1538-4357/ae2de8. arXiv:2509.08962.

[1] **Chichura, P. M.**, A. Rahlin, A. J. Anderson, et al. 2025. "Pointing Accuracy Improvements for the South Pole Telescope with Machine Learning." *Journal of Astronomical Instrumentation* 14 (January): 2550001. DOI: 10.1142/S2251171725500011. arXiv:2412.15167.

[16] Coerver, A., J. A. Zebrowski, S. Takakura, et al. 2025. "Measurement and Modeling of Polarized Atmosphere at the South Pole with SPT-3G." *ApJ* 982, no. 1 (March): 15. DOI: 10.3847/1538-4357/ada35d. arXiv:2407.20579.

[17] Event Horizon Telescope Collaboration et al. 2025. "Horizon-scale variability of M87* from 20172021 EHT observations." *A&A* 704 (December): A91. DOI: 10.1051/0004-6361/202555855. arXiv:2509.24593.

[18] Foster, A., A. Chokshi, A. J. Anderson, et al. 2025. "Detection of Thermal Emission at Millimeter Wavelengths from Low-Earth Orbit Satellites." *The Open Journal of Astrophysics* 8 (May): 51. DOI: 10.33232/001c.137526. arXiv:2411.03374.

[19] Ge, F., M. Millea, E. Camphuis, et al. 2025. "Cosmology from CMB lensing and delensed EE power spectra using 20192020 SPT-3G polarization data." *PhRvD* 111, no. 8 (April): 083534. DOI: 10.1103/PhysRevD.111.083534. arXiv:2411.06000.

[20] Khalife, A. R., L. Balkenhol, E. Camphuis, et al. 2025. "SPT-3G D1: Axion Early Dark Energy with CMB experiments and DESI." *arXiv e-prints* (July): arXiv:2507.23355. DOI: 10.48550/arXiv.2507.23355. arXiv:2507.23355.

[21] Kornoelje, K., L. E. Bleem, E. S. Rykoff, et al. 2025. "The SPT-Deep Cluster Catalog: Sunyaev-Zel'dovich Selected Clusters from Combined SPT-3G and SPTpol Measurements over 100 Square Degrees." *arXiv e-prints* (March): arXiv:2503.17271. DOI: 10.48550/arXiv.2503.17271. arXiv:2503.17271.

[22] Vitrier, A., K. Fichman, L. Balkenhol, et al. 2025. "Towards constraining cosmological parameters with SPT-3G observations of 25% of the sky." *arXiv e-prints* (October): arXiv:2510.24669. DOI: 10.48550/arXiv.2510.24669. arXiv:2510.24669.

[23] Zebrowski, J. A., C. L. Reichardt, A. J. Anderson, et al. 2025. "SPT-3G D1: Constraints on inflationary gravitational waves with two years of SPT-3G data." *PhRvD* 112, no. 12 (December): 123520. DOI: 10.1103/33wm-c6pt. arXiv:2505.02827.

[24] Ansarinejad, B., S. Raghunathan, T. M. C. Abbott, et al. 2024. "Mass calibration of DES Year-3 clusters via SPT-3G CMB cluster lensing." *JCAP* 2024, no. 7 (July): 024. DOI: 10.1088/1475-7516/2024/07/024. arXiv:2404.02153.

[25] Prabhu, K., S. Raghunathan, M. Millea, et al. 2024. "Testing the ΛCDM Cosmological Model with Forthcoming Measurements of the Cosmic Microwave Background with SPT-3G." *ApJ* 973, no. 1 (September): 4. DOI: 10.3847/1538-4357/ad5ff1. arXiv:2403.17925.

[26] Raghunathan, S., P. A. R. Ade, A. J. Anderson, et al. 2024. "First Constraints on the Epoch of Reionization Using the Non-Gaussianity of the Kinematic Sunyaev-Zel'dovich Effect from the South Pole Telescope and Herschel-SPIRE Observations." *PhRvL* 133, no. 12 (September): 121004. DOI: 10.1103/PhysRevLett.133.121004. arXiv:2403.02337.

[27] Tandoi, C., S. Guns, A. Foster, et al. 2024. "Flaring Stars in a Nontargeted Millimeter-wave Survey with SPT-3G." *ApJ* 972, no. 1 (September): 6. DOI: 10.3847/1538-4357/ad58db. arXiv:2401.13525.

[28] Balkenhol, L., D. Dutcher, A. Spurio Mancini, et al. 2023. "Measurement of the CMB temperature power spectrum and constraints on cosmology from the SPT-3G 2018 TT, TE, and EE dataset." *PhRvD* 108, no. 2 (July): 023510. DOI: 10.1103/PhysRevD.108.023510. arXiv:2212.05642.

[29] Pan, Z., F. Bianchini, W. L. K. Wu, et al. 2023. "Measurement of gravitational lensing of the cosmic microwave background using SPT-3G 2018 data." *PhRvD* 108, no. 12 (December): 122005. DOI: 10.1103/PhysRevD.108.122005. arXiv:2308.11608.

[30] Schiappucci, E., F. Bianchini, M. Aguena, et al. 2023. "Measurement of the mean central optical depth of galaxy clusters via the pairwise kinematic Sunyaev-Zel'dovich effect with SPT-3G and DES." *PhRvD* 107, no. 4 (February): 042004. DOI: 10.1103/PhysRevD.107.042004. arXiv:2207.11937.

[2] **Chichura, P. M.**, A. Foster, C. Patel, et al. 2022. "Asteroid Measurements at Millimeter Wavelengths with the South Pole Telescope." *ApJ* 936, no. 2 (September): 173. DOI: 10.3847/1538-4357/ac89ec. arXiv:2202.01406.

[31] Ferguson, K. R., A. J. Anderson, N. Whitehorn, et al. 2022. "Searching for axionlike time-dependent cosmic birefringence with data from SPT-3G." *PhRvD* 106, no. 4 (August): 042011. DOI: 10.1103/PhysRevD.106.042011. arXiv:2203.16567.

[32] Ghosh, A., F. Mertens, G. Bernardi, et al. 2020. "Foreground modelling via Gaussian process regression: an application to HERA data." *MNRAS* 495, no. 3 (January): 2813–2826. DOI: 10.1093/mnras/staa1331. arXiv:2004.06041.

[33] Kohn, S. A., J. E. Aguirre, P. La Plante, et al. 2019. "The HERA-19 Commissioning Array: Direction-dependent Effects." *ApJ* 882, no. 1 (September): 58. DOI: 10.3847/1538-4357/ab2f72. arXiv:1802.04151.